

RESEARCH ARTICLE

Analysis of Occupational and Non-Occupational Factors Related to Spinal Pain in Office Workers Who Use Computer: A Cross-Sectional Study From Türkiye

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Abstract: *Background:* Identifying work-related musculoskeletal pain and the factors that influence it is the first step in preventing and reducing office worker complaints. *Purpose:* This study aimed to analyze occupational and non-occupational risk factors related to spinal pain and to develop a model for predicting the risk of spinal pain in office workers. *Methods:* A cross-sectional study design was used. The Cornell Musculoskeletal Discomfort Questionnaire was used to determine the presence of spinal pain. The sample consisted of computer-using office workers from a university ($N=264$), of whom 162 (61.3%) were male, 102 (38.7%) were female. They were assessed for occupational (work-related inquiries, Rapid Upper Limb Assessment [RULA] evaluates the risk of workers to upper extremity-related ergonomic risk factors, and Rapid Office Strain Assessment [ROSA] evaluates workplace-related ergonomic risk factors) and non-occupational factors (demographic information). Independent samples *t*-test was used to compare office workers with and without spinal pain. The model was developed using stepwise multiple regression analysis. *Results:* Chair features were linked to pain in the neck, upper and lower back. Computer peripherals were related to the pain in the neck and upper back. Accurate predictors of spinal pain in computer-using office workers were identified as older age, female gender, higher BMI, prolonged weekly computer use time, no passive rest breaks, higher RULA and ROSA scores. *Conclusion:* Prevention of spinal pain should be focused on comprehensive health strategies aimed at increasing awareness of work posture and workplace ergonomics, especially for older, female, and high BMI computer-using office workers.

Keywords: musculoskeletal disorders, office ergonomics, occupational health, neck pain, back pain

Background

Spinal pain that consists of the neck, upper back, and low back pain is a chronic problem characterized by recurrent episodes (De Campos & Elkins, 2021). Spinal pain is a prevalent problem among computer-using office workers, with 12-month prevalence rates of 42% to 63% for neck pain, 66% for upper back pain, and 55% for lower back pain (Ardahan & Simsek, 2016; Bin Ahmed et al., 2023; Chen et al., 2018).

Office workers have frequently experienced sedentary tasks such as computer use, reading, presentations, telephone calls, etc. These activities necessitate prolonged periods of sitting, leading office workers to maintain a static work posture, which may be a risk factor for musculoskeletal pain due to postural malalignment (So et al., 2019). In addition, office workers with non-ergonomic working conditions, such as a lack of proper back support or a forward-leaning posture with kyphotic curvature, have more musculoskeletal pain in the spine than those who follow ergonomic working practices (Akkarakittichoke et al., 2022; Basaklı Calık et al., 2022; James et al., 2018).

There has been some attention in the literature regarding the relationship between individual factors, and physical factors of workplace and the development of spinal pain in the office workers. Inadequate chair features, such as lacking armrests and non-adjustable backrests, as well as improper positioning of computer devices, including mouse, keyboard, and monitor, have been associated with increased reports of spinal pain among office workers as workplace factors (Argus & Paasuke,

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The results of this study shed light on the associations between spinal pain and various occupational and non-occupational risk factors. In particular, older age, female gender, high BMI, prolonged weekly computer use, no passive rest break, and worse RULA and ROSA scores as significant predictors of the presence of spinal pain in office workers who use computers. Future studies could explore alternative methodologies (e.g., qualitative, mixed methods) to deepen our understanding of how these factors affect spinal pain among office workers of university staff. Longitudinal studies may also provide valuable insights into the effectiveness of health strategies in preventing and reducing spinal pain over time. Such research efforts hold great promise for advancing occupational health and promoting a supportive and flourishing work environment for computer-using office workers.

2022; Channak et al., 2022; Nunes et al., 2021). Factors such as longer sitting time, working posture, age, and gender are examples of individual factors that are often emphasized (Dong et al., 2021; Janwantanakul et al., 2012; Park et al., 2023).

Several studies conducted among office workers have reported a high prevalence of spinal pain (Nunes et al., 2021; Oliv et al., 2019; Russo et al., 2021), with evidence suggesting that its prevalence is gradually increasing and leading to both socio-economic and personal consequences (GBD 2021 Low Back Pain Collaborators, 2023; GBD 2021 Neck Pain Collaborators, 2024). Therefore, musculoskeletal pain is recognized as a significant work-related health issue in office workers; however, the most effective treatment for this group remains unknown (Eisele-Metzger et al., 2023). It is considered that the best intervention to decrease symptoms of work-related musculoskeletal pain is the prevention of risk factors (Buckle, 2005), the first step of which is analyzing related risk factors (Kaya Aytutuldu et al., 2022).

Office workers commonly complain of pain that affects more than one area of the spine rather than just a single area (Sihawong et al., 2016). Therefore, identifying factors that expose multiple regions to risk, rather than focusing solely on risk factors for a single region within this group, may offer a more comprehensive approach to developing strategies for reducing pain. Different risk factors for neck, upper back, and lower back have already been described in previous studies. As is evident from the literature, age, working without break, screen location, and use of a mouse have been reported for neck pain risk factors (Nunes et al., 2021). Additionally, age, non-ergonomic workstation, and prolonged computer time are also examples of reported risk factors for low back pain (Malińska et al., 2021). On the other hand, the upper back, along with the lower back and neck, is one of the most commonly reported areas of work-related musculoskeletal pain

(Demissie et al., 2024). However, it is noteworthy that there is a lack of research on risk factors associated with neck pain in the office worker population.

A model to predict spinal pain risk in office workers who use computers could help identify and reduce work-related spinal pain risks. Literature considering such predictive models is scarce in this population. Studies have been conducted in countries such as India, New Zealand, and Europe; however, their results may not be generalized to because of differences in socio-cultural, economic, and healthcare systems across nations (Jun et al., 2021; Malińska et al., 2021; Nunes et al., 2021; Sasikumar & Binoosh, 2020). Hence, the objectives of our study were to analyze the occupational and non-occupational risk factors related to spinal pain and to develop a model for predicting the risk of spinal pain in office workers who use computers in Türkiye. We hypothesized that there are differences in occupational and non-occupational factors between office workers with spinal pain and those of without spinal pain. We also hypothesized that occupational and non-occupational factors may be risk factors for the experience of spinal pain in office workers.

Methods

This was a descriptive-analytical, cross-sectional, and case-control study. Data were collected through face-to-face assessments, including ergonomic evaluations and self-reported questionnaires, focusing on variables related to individual and work factors, working posture, ergonomics, and musculoskeletal discomfort.

Participants

Two hundred sixty-four ($N=264$) office workers who regularly use computers in Muğla Sıtkı Koçman University were included in the study. Participants consisted of employees who work in Muğla Sıtkı Koçman University as academic and administrative staff. Participants were categorized into two groups based on their reported spinal pain: the spinal pain group and the no spinal pain group.

Inclusion criteria for all participants were as follows: (1) aged between 18 and 65 years, (2) an office worker who occupied the current office for at least 1 year, (3) computer-user at least 4 hours in a daily work routine, (4) volunteer to participate in the study. In addition to these inclusion criteria, reporting persistent pain in one or more of the neck, and upper/lower back regions for at least 3 months was an inclusion criterion for the spinal pain group. On the other hand, the absence of pain in any of the neck, and upper/lower back regions for the last 3 months was an inclusion criterion for the no spinal pain group. The exclusion criteria for all participants were as follows: presence of metabolic or inflammatory conditions, presence of infections, take medication or drugs during the assessment day, history of surgery for the head, neck, spine, and extremities, presence of vision and hearing problems, or pregnancy.

Measures

All assessments were conducted face-to-face by three intern physical therapists who had been trained by the senior researcher. The senior researcher, who had 30 years of ergonomic assessment experience, provided practical training to standardize the assessments of the three physical therapists. Twenty-five office workers whose data were not included in the study and determined by simple randomization were evaluated independently and blinded by physical therapists. Analyses of their assessment scores showed that the interclass correlation coefficient (ICC) of the assessors was greater than .90, indicating that there was standardization among assessors.

In order to determine risk factors related to the individual and work, participants were questioned about their work-related factors and habits, regarding the following factors: age, gender, body mass index (BMI, kg/m²), job tenure (years), work duration (hours/day), daily computer use time (hours/day), weekly computer use time (hours/week), passive rest breaks in working hours (yes/no), active rest breaks during working hours (yes/no).

Rapid Upper Limb Assessment (RULA), which scores the working posture of workers, was used to evaluate the exposure of workers to upper extremity-related ergonomic risk factors. This ergonomic assessment tool primarily evaluates posture and risk factors of the upper extremities and briefly considers the lower extremities, trunk, and neck according to the postural load of specific job tasks. A total score is calculated from the scores of upper extremity, trunk, and lower extremity, and measure of risk sections. A total score also from 1 to 7 and a higher score indicates a greater risk of ergonomic hazard (McAtamney & Nigel Corlett, 1993).

The rapid office strain assessment (ROSA) was used to assess the workplace to identify workplace-related ergonomic risk factors. ROSA allows for assessment of the chair, monitor and telephone, and keyboard and mouse in office workers who use computers. ROSA scores based on three sections: section A (the chair), section B (screen and phone), and section C (keyboard and mouse). Total ROSA scores ranged from 1 to 10, and an increase in the score indicates an increased presence of risk factors (Sonne et al., 2012).

The Cornell Musculoskeletal Discomfort Questionnaire (CMDQ) was used to determine the frequency and severity of work-related body discomfort during the last week (Cornell University Ergonomics Web, <https://www.ergo.human.cornell.edu>). The CMDQ consists of three sections: frequency of discomfort, severity of discomfort, and impact of discomfort on work ability. A weighted score was calculated for each body part. The scores of neck, upper back, and lower back sections and their weighted scores were used for data analysis in the current study. A higher score indicates a higher level of discomfort. The Turkish validity and reliability of the CMDQ have been confirmed in a previous study (Cronbach's α = .88; Erdinc et al., 2011).

Data Analysis

The statistical population of the study consisted of all staff working in the field of academics and administration at Muğla Sıtkı Koçman University (3,007 employees). The sample size was calculated as 135 by G-Power—version of Mac OS 3.1 (type 1 error: 0.05, power: 85%, number of predictors: 11). The sample size was selected from all academics and administration personnel using a simple random sampling method.

The data were analyzed using SPSS version 22.0. The Kolmogorov-Smirnov test was used to test the normality of data distribution. Demographic variables were compared between the groups using an Independent Samples *t*-test for continuous variables and a χ^2 test for categorical data. Pearson correlation coefficient (*r*) was used to test the strength of association among CMDQ, RULA, and ROSA. The “*r*” value was classified as low (.20–.39), moderate (.40–.59), strong (.60–.79), and very strong (.80–1.00). We used a binary logistic regression model in order to predict factors that represent spinal pain. Age, gender, BMI (kg/m²), work duration, job tenure, daily computer use, weekly computer use, passive rest breaks, active rest breaks, RULA, and ROSA were considered independent variables. In the model, only significant variables of interest at a 20% threshold ($p < .2$) were introduced. The statistical significance level was defined as $p < .05$.

The Human Research Ethic Committee of Muğla Sıtkı Koçman University with the decision number 210052/49 approved this research.

Results

A total of 264 participants from the original study met the criteria for inclusion in our analysis. Of these, 156 had spinal pain and 108 had no spinal pain. Male office workers constituted the majority in both the spinal pain group ($n=93$, 59.7%) and the no spinal pain group ($n=69$, 63.9%). Occupational and non-occupational characteristics of the study population are presented in Table 1. Participants with spinal pain had significantly higher job tenure, daily computer use, and weekly computer use time than participants without pain ($p=.016$, .027, .031, respectively). There was no statistical significance between groups regarding gender, passive rest breaks, and active rest breaks ($p>.05$).

Table 2 presents the frequency of reported musculoskeletal disorder symptoms in different spinal regions among the Spinal Pain Group during the last 12 months. Upper back was the area most frequently reported by office workers with spinal pain (87.8%) followed by neck (79.4%) and lower back (62.8%).

Table 3 demonstrates the between group differences in ergonomic factors, as measured by RULA and ROSA. The results showed that office workers with spinal pain had significantly higher RULA ($M=4.5$, $SD=0.9$) and ROSA ($M=4.3$, $SD=1.5$) scores compared to those without pain ($M=3.3$, $SD=1.1$; $M=2.3$, $SD=0.9$, respectively; $p=.020$ and $p<.001$). Moreover, all sections of ROSA scores were greater in the spinal pain group than those no pain group ($p<.001$).

Table 1. Occupational and Non-Occupational Characteristics of the Study Population According to the Presence of Pain

Variable	Spinal pain (<i>n</i> = 156)	No pain (<i>n</i> = 108)	<i>p</i> -Value
Age	42.9 (9.6)	41.8 (10.4)	.018*
Gender			.284
Female	63 (40.3%)	39 (36.1%)	
Male	93 (59.7%)	69 (63.9%)	
BMI (kg/m ²)	26.5 (4.4)	25.1 (3.6)	.307
Job tenure (years)	17.4 (9.9)	15.5(10.7)	.016*
Work duration (hours/day)	8.3 (1.5)	8.3 (1.5)	.441
Daily computer use (hours)	6.2 (1.9)	5.1 (2.7)	.027*
Weekly computer use (hours)	35.7 (13.0)	28.3 (15.3)	.031*
Passive rest breaks			.242
Yes	132 (84.6%)	87 (80.5%)	
No	24 (15.4%)	21 (19.5%)	
Active rest breaks			.056
Yes	33 (21.1%)	32 (29.7%)	
No	123 (78.9%)	76 (70.3%)	

Note. Data presented as Mean (Standard Deviation) or number (percentage); BMI = body mass index.

**p* < .05.

Table 2. Frequency of Reported Musculoskeletal Disorder Symptoms in Different Spinal Regions Among Spinal Pain Group During the Last 12 Months

Spinal region	CMDQ (<i>n</i> = 156)	
	N (%)	Severity
Neck	137 (87.8)	2.1 (2.2)
Upper back	124 (79.4)	2.2 (2.6)
Lower back	98 (62.8)	1.7 (1.6)

Note. Data presented as Mean (Standard Deviation) or number (percentage).

Correlations between Cornell Musculoskeletal Disability Questionnaire scores of spinal regions and RULA and ROSA scores are presented in Table 4 for the spinal pain group. Both RULA ($r = .734$, $p < .001$) and total ROSA ($r = .719$, $p < .001$) scores are strongly correlated with weighted CMDQ scores. Furthermore, we examined the correlations between sections of

ROSA scores and CMDQ scores. Height, pan depth, armrest, and back support of the chair were moderately correlated with CMDQ scores of the neck ($r = .315$ – $.564$, $p < .001$) and moderately to strongly correlated with upper back ($r = .430$ – $.625$, $p < .001$). Height and pan depth demonstrated strong correlations with CMDQ scores for lower back pain ($r = .606$ – $.647$, $p < .001$), whereas armrest and back support displayed moderate correlations ($r = .418$ – $.471$, $p < .001$). There was a strong correlation between monitor and CMDQ scores in the neck and upper back ($r = .607$ – $.668$, $p < .001$); conversely, there was a weak correlation with the lower back ($r = .206$ – $.248$, $p < .001$). The mouse section showed a moderate to strong correlation with the neck and upper back ($r = .415$ and $.653$, $p < .001$) and a weak correlation with the lower back ($r = .103$ – $.118$, $p < .001$). The keyboard section showed a strong correlation with the neck and upper back ($r = .601$ – $.673$, $p < .001$) and a weak correlation with the lower back ($r = .107$ – $.114$, $p < .001$).

Binary logistic regression analysis was performed using the backward stepwise method. The final model was obtained in the fifth step. The final model adopted age, gender, BMI, weekly computer usage time, rest breaks in working hours, RULA score, and ROSA score but excluded daily working hours,

Table 3. Ergonomic Analysis Between Groups

Variable	Spinal pain (<i>n</i> = 156)	No pain (<i>n</i> = 108)	Effect size (<i>d</i>)	<i>p</i> -Value
RULA	4.5 (0.9)	3.3 (1.1)	1.15	.020*
ROSA	4.3 (1.5)	2.3 (0.9)	1.55	<.001*
Section A—Chair	2.1 (1.6)	1.2 (1.2)	0.62	<.001*
Height and pan depth	1.6 (0.9)	0.8 (0.7)	0.90	<.001*
Armrest and back support	2.1 (1.7)	0.9 (0.6)	0.87	<.001*
Section B—Monitor and telephone	3.4 (1.6)	2.1 (1.0)	1	<.001*
Monitor	3.1 (1.9)	2.0 (0.9)	0.77	<.001*
Phone	1.5 (1.1)	1.1 (0.8)	0.45	<.001*
Section C—Mouse and keyboard	4.2 (1.5)	2.2 (1.7)	1.20	<.001*
Mouse	2.2 (1.1)	1.5 (0.8)	0.72	<.001*
Keyboard	3.4 (1.2)	1.7 (0.6)	1.76	<.001*

Note. Data presented as Mean (Standard Deviation). RULA = rapid upper limb assessment; ROSA = rapid office strain assessment.

**p* < .05.

working years, daily computer usage time, and exercise during work hours ($\chi^2(8) = 14.722$, $p = .065$; Table 5). The model correctly predicted 79.5% of cases who have spinal pain and 67.6% of subjects who have no spinal pain, giving an overall percentage correct prediction rate of 74.6%.

The results of odds ratio (*OR*) identified age ($OR = 0.944$; 95% CI [0.91, 0.98]), female ($OR = 0.330$; 95% CI [0.14, 0.76]), BMI ($OR = 0.836$; 95% CI [0.75, 0.92]), weekly computer usage time ($OR = 0.956$; 95% CI [0.93, 0.98]), passive rest breaks ($OR = 0.432$, 95% CI [0.17, 1.04]), RULA score ($OR = 0.363$; 95% CI [0.25, 0.51]), and ROSA score ($OR = 0.491$; 95% CI [0.35, 0.68]) as risk factors for experiencing pain with a statistically significant level (Table 5).

Discussion

The current study presents a model for predicting spinal pain exposure in office workers. The results of our study suggest that age, gender, BMI, weekly computer use time, passive rest breaks, RULA score, and ROSA score are accurate predictors of spinal pain. Hence, the hypothesis assuming that occupational and non-occupational factors may be risk factors for the experience of spinal pain in office workers was confirmed. Our results also suggest that office workers with spinal pain had worse work postures and were exposed to more risk factors in the office work environment than those without pain. Therefore, the hypothesis assuming that there are

differences in occupational and non-occupational factors between office workers with spinal pain and those without spinal pain was confirmed. Our model may assist occupational therapists, physical therapists, and other professionals in assessing spinal pain and deciding on interventions to prevent spinal pain.

Musculoskeletal Symptoms

In our study, most of the office workers with spinal pain reported neck and upper back pain, respectively. On the other hand, although the lower back was the least reported pain area, it still had a 62.8% reporting rate. Similarly, James et al. (2018) reported a 70% reporting rate for neck and upper back pain and 60% for lower back pain in the office workers who were employed in the university setting. In another study conducted with office workers employed at a university, upper back, neck, and lower back areas were reported as the most painful areas, respectively (Basakci Calik et al., 2022).

Our model presents some individual and work-related factors as indicators of spinal pain in office workers. Sasikumar and Binoosh (2020) have developed a model for predicting the risk of musculoskeletal disorders among computer professionals. Similar to our findings, age, gender, height, and weight were each identified as factors influencing pain in spinal regions, including the neck, upper back, and lower back. Moreover, they also reported job tenure and work duration as predictors. On

Table 4. Correlations Between Cornell Musculoskeletal Disability Questionnaire Scores and RULA and ROSA Scores

CMDQ		Weighted score (r)	Neck frequency	Neck severity	Neck work interference	Upper back frequency	Upper back severity	Upper back work interference	Lower back frequency	Lower back severity	Lower back work interference
	RULA	.734**	.594**	.490**	.413**	.440**	.418**	.430**	.509**	.596**	.486**
	ROSA	.719**	.522**	.485*	.413**	.445**	.412**	.552**	.637**	.687**	.622**
	Height and pan depth	.532**	.321**	.436**	.315**	.430**	.425**	.442**	.625**	.606**	.647**
	Armrest and back support	.564**	.441**	.408*	.478**	.567**	.546**	.539**	.418**	.448**	.471**
	Monitor	.668**	.641**	.607**	.615**	.627**	.668**	.629**	.206**	.212**	.248**
	Phone	.245**	.205**	.214**	.208**	.261**	.259**	.217**	.107**	.111**	.121**
	Mouse	.653**	.415**	.484**	.474**	.524**	.576**	.504**	.118**	.103**	.116**
	Keyboard	.638**	.612**	.685**	.601**	.655**	.601**	.673**	.108**	.114**	.107**

Note. Spearman's correlation test (r); RULA = rapid upper limb assessment; ROSA = rapid office strain assessment.

* $p < .05$. ** $p < .001$.

Table 5. Results of Backward Stepwise Regression Analyses

Dependent variable	Independent variables analyzed	B	SE	Wald	p-Value	OR	95% confidence interval of the difference	
							Lower	Upper
Presence of pain	Constant	15.307	2.407	40.456	<.001*			
	Age	-0.057	0.019	9.056	.003*	0.944	0.910	0.980
	Gender	-1.108	0.431	6.604	.010*	0.330	0.142	0.769
	BMI (kg/m ²)	-0.180	0.049	13.366	<.001*	0.836	0.759	0.920
	Weekly computer use (hour)	-0.045	0.013	12.326	<.001*	0.956	0.933	0.980
	Passive rest breaks	-0.839	0.450	3.479	.062	0.432	0.179	1.044
	RULA	-1.013	0.181	31.333	<.001*	0.363	0.255	0.518
	ROSA	-0.712	0.172	17.203	<.001*	0.491	0.351	0.687

Note. B= Estimated value of the regression coefficient; SE= Standard error; Wald = B/SE; OR= odds ratio; RULA = rapid upper limb assessment; ROSA = rapid office strain assessment; BMI = body mass index.

* $p < .05$.

the other hand, work breaks did not have an association with musculoskeletal symptoms in their study population. In another study, older age, female gender, increased sitting hours, higher job strain, poor working posture, cervical range of motion, and muscular endurance were indicated as risk factors for neck pain in office workers (Jun et al., 2021). Further, rest breaks, backrest and armrest, distance between mouse and keyboard, working posture of neck above the 45° inclination, and working posture of forearms were reported as risk factors for lower back pain in the office workers (Celik et al., 2018).

Age, gender, and BMI were individual factors and reported as significant factors for musculoskeletal discomfort in previous studies (Bin Ahmed et al., 2023; Kaliniene et al., 2016; Sethi et al., 2011). There is conflicting evidence associated with age. Several studies did not find a relationship between age and spinal pain (Ranasinghe et al., 2011). However, some studies reported an increased risk with age (Nunes et al., 2021). A similar inconsistency is evident in the literature regarding sex distribution. Cho et al. (2012) reported sex differences in musculoskeletal symptoms in office workers, with male computer users having the highest prevalence of the neck area (74%), while female computer users had the highest prevalence for the shoulder region (83%). Conversely, Kaya Aytutuldu et al. (2022) and Rodrigues et al. (2017) indicated that sex made no difference in musculoskeletal symptoms. On the other hand, several studies have identified female gender as a potential risk

factor (Paksaichol et al., 2012). Height led workers to change their posture to adapt to their workplace (Marcus et al., 2002). If workers cannot make an ergonomic adjustment, they may experience musculoskeletal disorders. Similarly, weight contributes to the physiological and mechanical overload of tissues. Prolonged sitting in front of a computer increases tissue stress, and this effect may be amplified in individuals with a high BMI, further increasing the risk of such disorders (Sethi et al., 2011). Our results, which identified BMI as a risk factor for spinal pain in office workers, were similar to this report.

Weekly computer usage time, rest breaks during working, RULA score, and ROSA score were ergonomic factors in our statistical model. Spending more than 7 hours a day in front of the computer was reported as a highly correlated factor for musculoskeletal pain in all regions of the body (Ardahan & Simsek, 2016). Spyropoulos et al. (2007) indicated that using a computer over 6 hours a day was a related factor for low back pain. In another study, more than 40 hours a week was noted as a risk factor for neck problems in women and men (Malińska et al., 2021). Based on our results, office workers with spinal pain had statistically more daily and weekly computer use.

Our regression model included only weekly computer use for presence of spinal pain and we reported that every 1-hour increase in weekly computer use increases the risk of spinal pain in office workers by 0.95 times.

Rest breaks were recommended for decreasing adverse effects of prolonged sitting with poor posture. Several studies have reported the effectiveness of breaks in reducing spinal pain, including neck and back (Galinsky et al., 2007; Sheahan et al., 2016; Waongenngarm et al., 2018). On the contrary, a meta-analysis study has indicated that breaks alone may be insufficient for office workers experiencing pain symptoms, and workers may derive greater benefits from a combination of breaks and exercise (Chen et al., 2018). Parallel to this report, in the present study, there is no difference between office workers with and without spinal pain regarding passive breaks; at least 80% of workers in both groups engaged in passive breaks. In addition, the groups also did not show a difference regarding active breaks; the percentage of workers taking active breaks was less than 30%. Indeed, our results indicated that office workers frequently engaged in passive breaks but exhibit reluctance toward taking active breaks. This finding might emphasize the importance of strategies that increase active breaks for future studies in office workers.

Previous research has reported the association of work duration and job tenure with musculoskeletal disorders in office workers using computers (Sharan & Ajeesh, 2012). According to our findings, while there was a significant difference in the job tenure between the groups, there was no significant difference in the work duration. This is an expected result because our research population consisted of office workers with the same work duration but different job tenures. Sasikumar and Binoosh (2020) reported a significant correlation between job tenure and neck and lower back pain, similar to our results. However, although job tenure was significantly higher in the spinal pain group, neither job tenure nor work duration were included in our final model. Nevertheless, the higher job tenure in one group compared to the other may have potentially contributed to the development of spinal pain. This factor should be considered when interpreting the results of the study.

Workplace Environment

In the current study, office workers with spinal pain had worse ROSA score than those without spinal pain. Moreover, both scores were determined to be risk factors for presence of spinal pain. Previous research reported the risk score of ROSA was greater than 5 (Sonne et al., 2012) or 6 (Rodrigues et al., 2019) points for musculoskeletal discomfort. Final ROSA scores of their study were 4.13 and 5.83, respectively, and they analyzed musculoskeletal discomfort for all body regions. Neck/head and upper/lower back were also the most reported painful areas in their results. As a parallel finding in our study, a final ROSA score was 4.33 in office workers with spinal pain.

Keyboard, height and pan depth of chair, armrest and back support, and height and placement of monitor were the most contributing factors to the final ROSA for the presence of spinal pain (effect sizes were 1.76, 0.90, 0.87, and 0.77, respectively). Similarly, Kaya Aytutuldu et al. (2022) have reported that chair height, arm/back rest, and keyboard and mouse position were

related factors with the musculoskeletal pain, including neck and lower back. Distance from the keyboard and the near edge of the desk has been reported as a risk factor for neck pain in the office workers (Paksaichol et al., 2012). According to the findings of another study, it has been suggested that an improper chair position may impact the positioning of the upper extremities, especially when reaching for the mouse and keyboard. This, in turn, can contribute to musculoskeletal disorders in the neck and lower back among office workers (Rodrigues et al., 2017). Similarly, in our study, the RULA score was worse in office workers with spinal pain than in those without pain. Many previous studies have reported that the backrest of a chair may reduce lower back discomfort in office workers. An individually adjustable chair back supports lumbar lordosis and may potentially reduce sitting-related lower back pain (Carcone & Keir, 2007; Channak et al., 2022; Makhous et al., 2009). In addition, the pan depth of a chair may reduce ischial pressure (O'Sullivan et al., 2012). The integration of suitable seat depth and lumbar support has also been associated with a decrease in lower back pain (Makhous et al., 2003). The lack of armrests of chairs, on the other hand, has frequently been linked to neck pain (Jun et al., 2017; Paksaichol et al., 2012). To summarize, a chair with adjustable height, backrest, and armrests can reduce the pressure on intervertebral disc and muscle activity of the neck and back, thus contributing to the reduction of office workers' complaints of spinal pain (Van Niekerk et al., 2012). Improper computer workstation and awkward working postures in front of a computer screen were related to musculoskeletal pain in office workers (Jun et al., 2017). The distance between body and computer screen, working posture, and computer usage time were reported as related factors (Spyropoulos et al., 2007). In addition to extended periods spent in front of a computer screen, Ricco et al. (2016) have highlighted the influence of ergonomic desks and chairs on musculoskeletal discomfort. The longer duration of static working postures and inappropriate working environment among office workers increases the load on the joints and muscles. Particularly, higher loading is reported in the upper trapezius muscle for keyboard users (Cooper & Straker, 1988), and it may be attributed as a possible cause of neck pain. On the other hand, back pain seems to be related to prolonged sitting and sustained kyphotic posture and lumbar flexion (Janwantanakul et al., 2012).

Working Posture

The neck and shoulders are the most reported areas for musculoskeletal disorders in office workers, especially in using computers. Therefore, upper and lower back disorders and their relationship with the working posture of the upper extremity were less investigated in office workers. On the other hand, our results, along with those of several studies, suggested the associations between working posture and not only neck pain but also upper and lower back pain (Kalinienė et al., 2016; Sasikumar & Binoosh, 2020). Frankly, the precise explanation for the relationship between back pain and working posture of

the upper extremity is unclear. The higher RULA scores have been frequently associated with neck and upper extremity pain in the office workers. The frequently cited explanation for neck pain is static posture, leading to an overload on the muscles and joints. In addition to these, prolonged use of the mouse and keyboard, along with their improper ergonomic placement, has been linked with upper extremity pain (Côté et al., 2008; Ijmker et al., 2007; Waersted et al., 2010). The longer duration of static working postures among office workers increases the load on the joints and muscles. Particularly, higher loading is reported in the upper trapezius muscle for keyboard users (Cooper & Straker, 1988), and it may be attributed as a possible cause of neck pain. On the other hand, Rodrigues et al. (2017) supposed that improper position of the chair can lead to awkward postures to reach the mouse and keyboard, consequently leading office workers to experience discomfort in the lower back and neck. In essence, the RULA evaluates the posture and ergonomic risk factors of the upper extremities. At the same time, it also provides a brief assessment of the lower extremities, trunk, and neck. Therefore, inappropriate trunk and neck positions in office workers with spinal pain are likely to have contributed to unfavorable RULA scores.

Earlier studies predominantly focused on analyzing occupational or non-occupational risk factors independently, while this study comprehensively integrated both dimensions allowing for a holistic understanding of the factors contributing to spinal pain. The current risk model not only identifies risk factors but also may offer practical implications for risk mitigation in the workplace. Furthermore, unlike previous research, this study addresses spinal pain encompassing both the back and lumbar regions, presenting a broader and more inclusive understanding of pain-related risk factors.

Limitations

This study has some limitations that should be highlighted. Firstly, the researchers had a limited timeframe for observing the participants and the observations took place within the participants' work environments during this restricted period. This circumstance may have led participants to involuntarily alter their posture. Therefore, further studies should consider an objective assessment method, such as video-based-3D observation, that covers a much longer period. Second, the results cannot be generalized to all office workers population because the study population consisted of computer users. Third, musculoskeletal symptoms were collected using a self-reporting method, which introduces the possibility of recall bias and may result in either overestimation or underestimation of the reported symptoms. Fourth, the study population had a wide age range, but age groups did not analyze data. In addition to these, there was a lack of information regarding the participants' habitual physical activity, which makes interpreting the results of this study difficult. Future research is warranted to investigate prevention strategies for spinal pain to decrease work-related musculoskeletal disorders in office workers.

Conclusion

This study suggested that office workers with spinal pain had worse ergonomic factors related to both work posture and workplace than those without spinal pain. Chair features, including armrest, back support, height, and pan depth, were linked to pain in the neck, upper back, and lower back. On the other hand, computer peripherals, including monitor, mouse, and keyboard were related to the pain in the neck and upper back. Older age, female gender, higher BMI, prolonged weekly computer use, no passive rest break, higher RULA score, and higher ROSA score were risk factors for the presence of spinal pain in office workers who use computers. Efforts to prevent spinal pain should prioritize comprehensive health strategies that promote awareness of proper work posture and workplace ergonomics, considering female gender, older age, and higher BMI among office workers who use computers.

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Author Contributions

All authors have contributions to the conception and design of the work, or analysis and interpretation of data for the work and drafting the article or revising it critically for important intellectual content. Finally, they approved the manuscript for publication and ensured that any questions related to the accuracy or integrity of the work were appropriately investigated and resolved.

Conflict of Interests

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Ethical Approval

Ethical approval has been provided by the Human Research Ethic Committee of Muğla Sıtkı Koçman University (12/29/2021, 210052/49).

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